

Jennifer Ren

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EDUCATION

B.S. in Electrical Engineering and Computer Sciences

UNIVERSITY OF CALIFORNIA, BERKELEY | GPA: 3.8/4.0

Aug 2024 – May 2028

Berkeley, CA

Coursework: Data Structures & Algorithms, Computer Architecture, Digital Design & IC, Design Automation, Operating Systems & Systems Programming, Machine Learning, Internet Protocols, Computer Security, Signals & Systems, Circuits & Devices, Discrete Math & Probability, Linear Algebra & Differential Equations, Multivariable Calculus

Interests: Graphic Design/Painting, Competitive Figure Skating, Poker

SKILLS

Languages SystemVerilog, Verilog, Python, C, C++, Java, x86/RISC-V Assembly, LaTeX, Go

EDA Tools Synopsys VCS, Synopsys Verdi, Cadence Genus, Cadence Innovus, Cadence Joules

Other Tools LTspice, Logisim, MATLAB, Oscilloscope, Spectrum Analyzer

PROJECTS

3-Stage Pipelined RISC-V CPU (EECS151)

SYSTEMVERILOG | SYNOPSYS VCS, VERDI | CADENCE GENUS, INNOVUS, JOULES

- Designed a 3-stage pipelined RISC-V CPU in SystemVerilog
- Implemented hazard detection, forwarding, stalls, and branch/jump flushing
- Verified with assembly tests and post-synthesis simulation; ran synthesis, place-and-route, and power analysis
- Placed 1st in the class leaderboard by frequency, with a top 10 figure of merit

2026

Semiconductor SEM Defect Classification (Carinthia)

PYTHON | PYTORCH | TIMM | COMPUTER VISION

- Benchmarked 4 vision models (Swin, Xception, TinyViT, EfficientFormer) for SEM defect classification in one shared PyTorch/timm multi-seed pipeline
- Best model reached 0.99 macro-F1; imbalance-aware training lifted rare-class recall (55 images, $\sim 73\times$ long-tail) to 100%

2026

EXPERIENCE

Undergraduate Researcher, AI-Driven Tapeout

UC BERKELEY EECS | ML FOR EDA | PROF. KARANDIKAR

- Using CHIA agent loops to optimize RTL design across the full RTL-to-GDSII tapeout flow
- Integrating Cadence Jasper formal verification into the group's AI-driven tapeout flow

Jun 2026 – Present

Berkeley, CA

Course Staff, EECS 151 (Digital Design & Integrated Circuits)

UC BERKELEY EECS DEPARTMENT

- UCS1 tutor for Berkeley's core digital design course covering SystemVerilog, CMOS, RTL design, and full ASIC/FPGA flow
- Assist in lab sections and hold office hours supporting students through ASIC-based RISC-V CPU design and synthesis labs

Aug 2026 – Present

Berkeley, CA

Machine Learning Intern

BFAI SEMICONDUCTOR | PYTHON, PYTORCH, COMPUTER VISION

- Developed ML pipelines for semiconductor defect detection using classification, detection, and segmentation models
- Built preprocessing and augmentation workflows for crack and void detection in Docker containers

Jan 2026 – Present

Coordinator, CSM66

COMPUTER SCIENCE MENTORS AT BERKELEY

- Organize content and mentor meetings to improve tutoring sessions resources for 200+ students
- Supported students in signals, linear algebra, and data processing concepts

Jan 2025 – Present

Berkeley, CA